

LABORATORY ACTIVITY Cell-to-Cell Communication

Name: Suan, Jade Angela B.

Date Completed: October 08, 2026

Sender Cell: Pituitary thyrotroph

Procedure

Part A. Choose and Register Your Sender Cell (10–15 min)

- **Sender cell:** Pituitary thyrotroph
- **Tissue/context:** Anterior pituitary gland. It regulates thyroid hormone production and metabolism through the hypothalamic–pituitary–thyroid axis.
- **Why it is a meaningful sender:** Thyrotrophs are the pituitary cells that control the thyroid gland. They release their signal into the blood, and it reaches the thyroid, which affects the body's metabolism.

Part B. Discover a Candidate Signal Produced by the Sender Cell (25–35 min)

ITEM	ANSWER
Ligand	TSH (CGA alpha subunit + TSHB beta subunit)
Evidence	In the Human Protein Atlas, TSHB shows cytoplasmic expression in a subset of endocrine cells in the anterior pituitary. Its RNA is tissue enriched in the pituitary gland, and the cell type classification lists pituitary thyrotropes as the enriched cell type. The protein is predicted to be secreted, with an extracellular location of "secreted to blood," and it is not detected in immune cells. HPA also lists enhanced RNA in spermatids in the single-cell data, but I consider the pituitary protein staining stronger evidence for the sender cell. This supports thyrotropes as the main source of TSHB.
Signaling type	Endocrine — TSH is released into the bloodstream and acts on distant thyroid follicular cells.

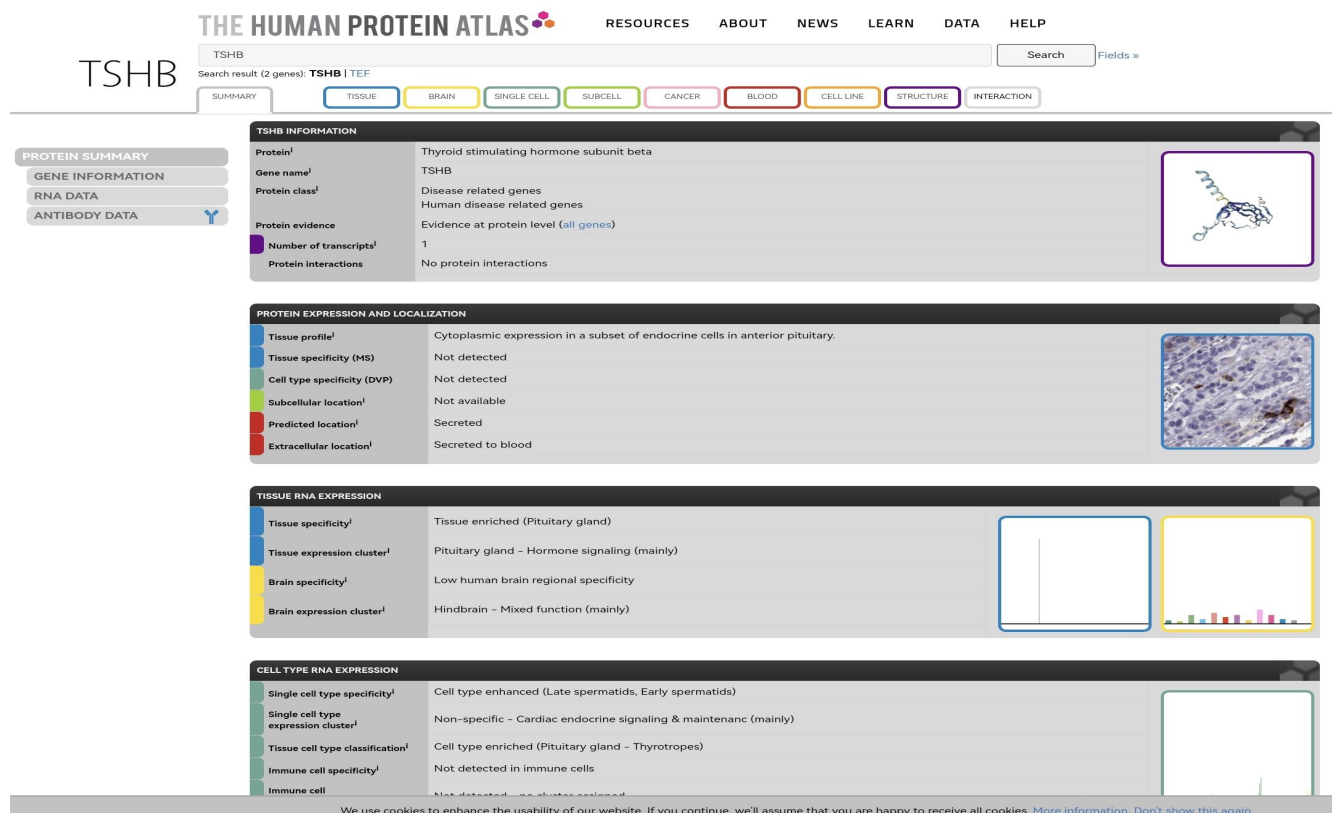


Figure 1. *Expression of TSHB in pituitary thyrotrophs – Human Protein Atlas*

Part C. Identify the Receptor and Receiver Cell (20–30 min)

ITEM	RECORDS
Ligand	TSH (CGA alpha + TSHB beta subunits)
Receptor	TSHR (Thyrotropin receptor), a G protein-coupled receptor
Receiver cell	Thyroid follicular cell (thyroid glandular cell in HPA)
Signaling Context	Regulation of thyroid hormone synthesis and metabolic homeostasis
Supporting source(s)	In the Human Protein Atlas, TSHR RNA is cell type enriched in thyroid glandular cells of the thyroid gland. UniProt describes TSHR as a G protein-coupled receptor for thyrotropin. In OmniPath Explorer, TSHB (P01222) is linked to TSHR (P16473) as a directed interaction supported by ligand-receptor resources (CellTalkDB, Cellinker, LKdb and connectomeDB2020).
Checkpoint	The pituitary thyrotroph makes TSH (TSHB + CGA), which can signal through TSHR on the thyroid follicular cell in endocrine signaling.

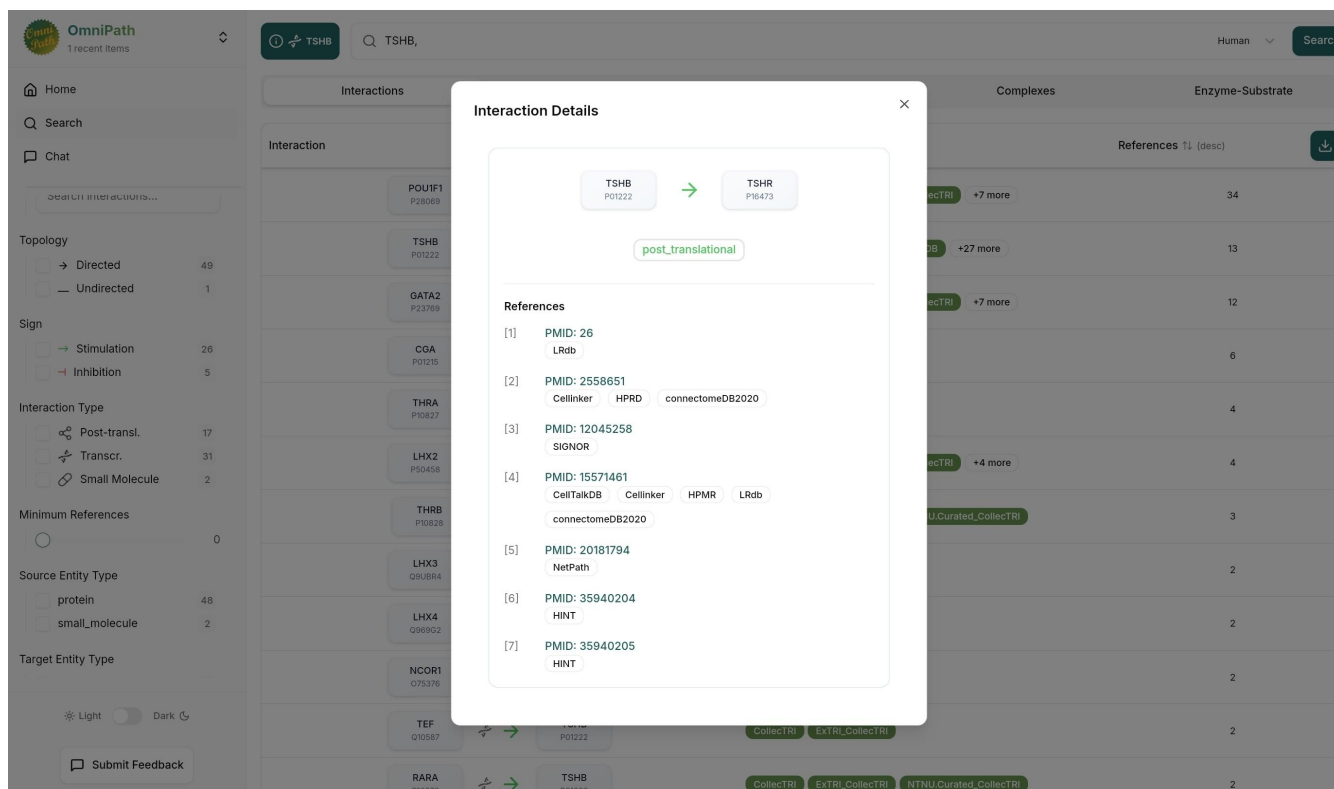


Figure 2. OmniPath Explorer record for the TSHB (TSH beta subunit) → TSHR interaction, supported by ligand-receptor resources such as CellTalkDB, Cellinker, LRdb and connectomeDB2020.

Part D. Explore the Receptor-Centered Network in STRING (25–35 min)

ITEM	RECORDS
Database	STRING v12.5
Organism	Homo sapiens
Input proteins	TSHR, TSHB, CGA, GNAS, PRKACA, CREB1, PAX8, TG, TPO
Network Statistics	9 nodes, 18 edges; expected number of edges: 2; PPI enrichment $p = 9.65e-12$
Enriched GO terms	GO:0006590 Thyroid hormone generation (3 of 16, FDR 0.00055); GO:0004996 Thyroid-stimulating hormone receptor activity (2 of 2, FDR 0.0040)
Enriched KEGG pathways	hsa04918 Thyroid hormone synthesis (8 of 73, FDR $1.88e-16$); hsa04024 cAMP signaling pathway (5 of 222, FDR $2.16e-06$)

Enriched Reactome/WikiPathways	HSA-418555 G alpha (s) signalling events (5 of 157, FDR 1.07e-05); WP2032 Thyroid stimulating hormone TSH signaling (5 of 58, FDR 1.13e-08)
Proteins linking receptor activation to the response	GNAS (Gs alpha, couples TSHR to cAMP); PRKACA (PKA catalytic subunit); CREB1 (cAMP-responsive transcription factor); PAX8 and TG (thyroid transcription factor and thyroglobulin)
Interpretation	STRING edges are functional associations, not proof of direct binding or direction. The order TSHR → GNAS → PRKACA → CREB1 comes from known cAMP/PKA biology, not from STRING.

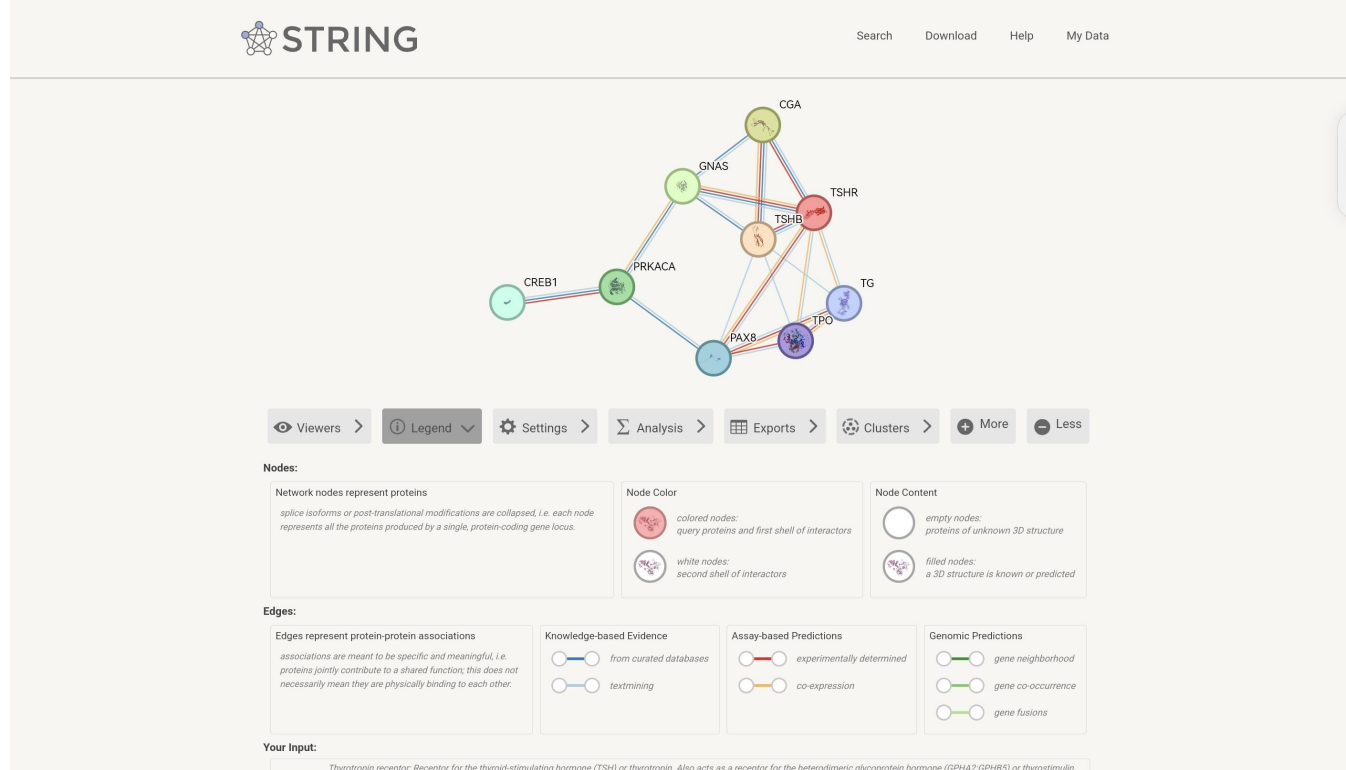


Figure 3. *STRING v12.5 functional association network (Homo sapiens) of TSHR, TSHB, CGA, GNAS, PRKACA, CREB1, PAX8, TG and TPO (9 nodes, 18 edges; PPI enrichment $p = 9.65e-12$).*

Part E. Validate One Molecular Interaction in IntAct (20–30 min)

ITEMS	RECORDS
Interacting molecules	PRKACA (UniProt P17612) and CREB1 (UniProt P16220), both proteins
Detection method	Anti-bait coimmunoprecipitation
Organism	<i>Homo sapiens</i> (both interactors)
Host organism	<i>Homo sapiens</i> HT-29 colorectal adenocarcinoma cells
Publication	PMID 37501933 (Fig. 4D)
Interaction type	Physical association
Interaction AC	EBI-46451554
Confidence (MI score)	0.4
Evidence type	Physical association in human cells. The record does not show direct binding, and it does not show that PRKACA phosphorylates CREB1.
Additional pairs	PRKACA-PRKAR1A (P17612, P10644) has multiple IntAct records, including a direct interaction detected in vitro by anti tag coimmunoprecipitation (EBI-16107177, PMID 24855271), plus phosphorylation, FRET and pull down records. PRKAR1A is not in my STRING network, so I used it only as a supporting pair. No IntAct record found for TSHR with TSHB.

Filters	Interactor Species	Interactor Type	Interaction Type	Interaction Detection Method	Interaction Host Organism	Mutation	Expansion	Positive	MI Score	Export	Network	Table
Select	Molecule A	Molecule B	Identifier A	Identifier B	Type A	Type B	Species A	Species B	Host Organism	Positive Interaction	Detection Method	Publication ID
☐	PRKACA	BAG2	UniProt P17612	UniProt Q95816	protein	protein	Homo sapiens	Homo sapiens	Homo sapiens HEK293T embryonic kidney cell	✓	anti bait coip	35167623
☐	PRKACA	BAG2	UniProt P17612	UniProt Q95816	protein	protein	Homo sapiens	Homo sapiens	Homo sapiens HEK293T embryonic kidney cell	✓	anti bait coip	35167623
☒	PRKACA	CREB1	UniProt P17612	UniProt P16220	protein	protein	Homo sapiens	Homo sapiens	Homo sapiens HT-29 colorectal adenocarcinoma cell	✓	anti bait coip	37501933
☐	PRKACA	SLTM	UniProt P17612	UniProt Q9NWH9	protein	protein	Homo sapiens	Homo sapiens	Homo sapiens U2OS osteosarcoma cell	✓	crosslink	10.1074/mcp.ra118.030021884
☐	PRKACA	SORBS3	UniProt P17612	UniProt Q60504-2	protein	protein	Homo sapiens	Homo sapiens	Chlorocebus aethiops	✓	anti tag coip	16483316 10.1111/j.1365-2443.2006.00909.x
☐	PRKACA	SRSF7	UniProt P17612	UniProt Q16629	protein	protein	Homo sapiens	Homo sapiens	Homo sapiens HEK293FT embryonic kidney cell	✓	anti tag coip	22677170
☐	RYR2	PRKACA	UniProt Q92736	UniProt P17612	protein	protein	Homo sapiens	Homo sapiens	Homo sapiens heart tissue	✓	anti bait coip	10830164
☐	WASF2	PRKACA	UniProt Q9Y6W5	UniProt P17612	protein	protein	Homo sapiens	Homo sapiens	Chlorocebus aethiops	✓	anti tag coip	16483316 10.1111/j.1365-2443.2006.00909.x
☐	CCDC88A	PRKACA	UniProt Q3V6T2	UniProt P17612	protein	protein	Homo sapiens	Homo sapiens	Saccharomyces cerevisiae	✓	two hybrid pooling	20936779

Figure 4. IntAct record of the PRKACA-CREB1 interaction (EBI-46451554), a physical association detected by anti-bait coimmunoprecipitation in human cells.

Part F. Build the Final Cell-to-Cell Communication Model (20–30 min)

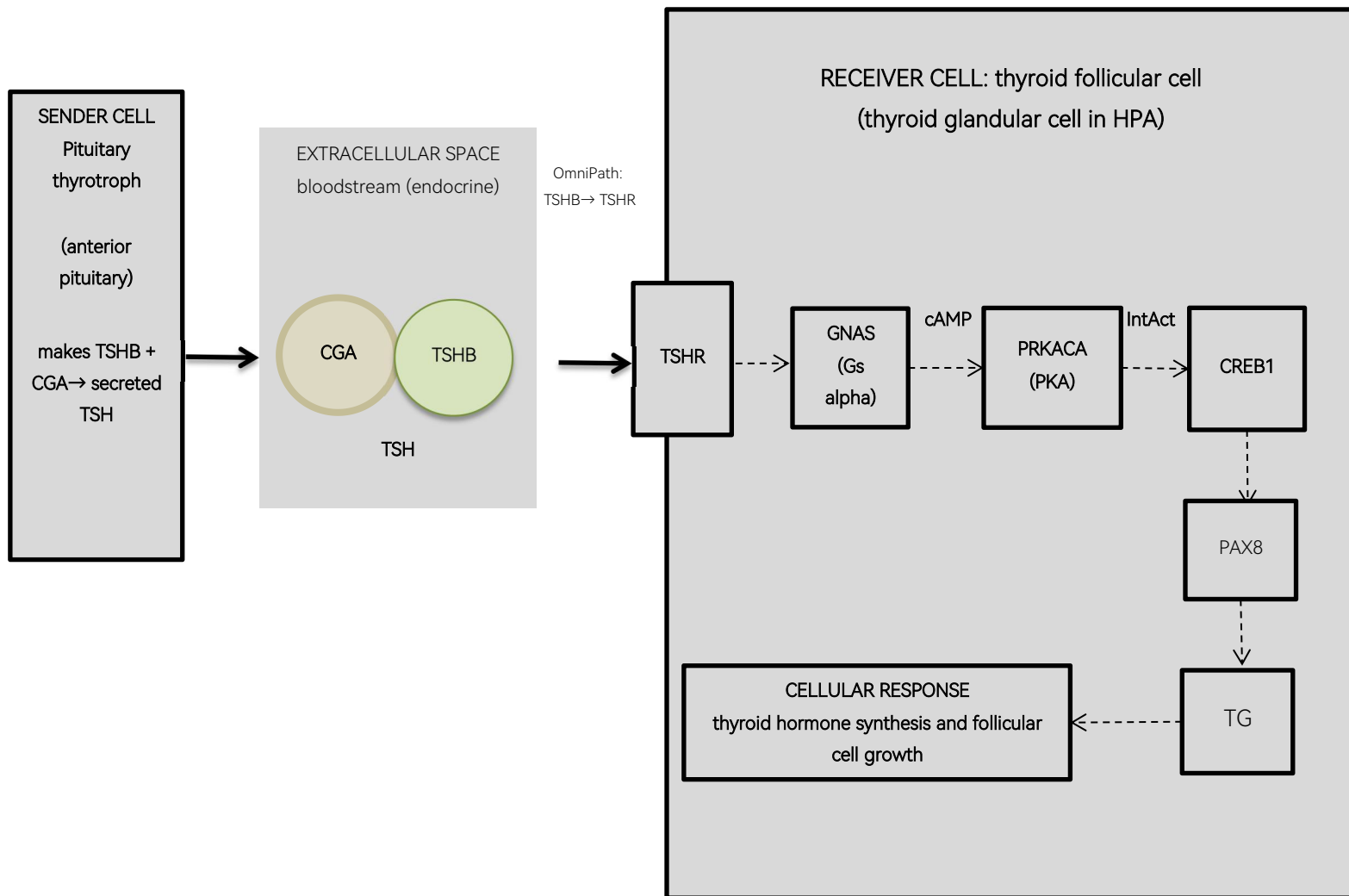


Figure 5. *Proposed model of endocrine signaling from the pituitary thyrotroph to the thyroid follicular cell through TSH and TSHR. Solid arrows are supported by database records; dashed arrows are inferred from known cAMP/PKA biology.*

Evidence used

HPA: TSHB in pituitary thyrotrophs; TSHR in thyroid glandular cells

OmniPath: TSHB → TSHR ligand-receptor record

STRING: 9 proteins, 18 edges (2 expected), $p = 9.65e-12$

IntAct: PRKACA-CREB1 physical association (EBI-46451554)

→ supported by database record -----> inferred from known cAMP/PKA biology

Interpretation:

I chose the pituitary thyrotroph as the sender cell. In the Human Protein Atlas, TSHB is enriched in the pituitary gland and in pituitary thyrotropes, and it is predicted to be secreted to the blood, so TSH (CGA + TSHB) acts as an endocrine signal. OmniPath links TSHB to TSHR (P01222 to P16473) and lists ligand-receptor resources such as CellTalkDB, Cellinker, LRdb and connectomeDB2020. HPA shows TSHR enriched in thyroid glandular cells, which supports the thyroid follicular cell as the receiver. In STRING, TSHR, TSHB, CGA, GNAS, PRKACA, CREB1, PAX8, TG and TPO formed a network with 18 edges instead of the 2 expected ($p = 9.65e-12$), enriched for thyroid hormone generation, cAMP signaling and TSH signaling. IntAct has a physical association record for PRKACA and CREB1 (EBI-46451554, anti-bait coimmunoprecipitation in human cells, MI score 0.4). The TSH to TSHR step and the thyroid-related network are the best supported parts. The order TSHR, GNAS, PRKACA, CREB1, PAX8 and TG is my inference from known cAMP/PKA biology, because STRING edges show association only and IntAct did not show direction or phosphorylation. The expected response is more thyroid hormone synthesis and follicular cell growth.

2. Required Evidence to Record

ITEMS	RECORDS
Sender cell	Pituitary thyrotroph (anterior pituitary gland). Context: regulation of thyroid hormone production and metabolism through the hypothalamic-pituitary-thyroid axis. It is a meaningful sender because thyrotrophs are the pituitary cells that stimulate the thyroid gland.
Signal/ligand	TSH (thyroid-stimulating hormone), made of the CGA (alpha) and TSHB (beta) subunits. HPA: TSHB is tissue enriched in the pituitary gland and cell type enriched in pituitary thyrotropes, with cytoplasmic expression in a subset of endocrine cells in the anterior pituitary. Predicted secreted, “secreted to blood” (endocrine signaling).
Receptor	TSHR (thyrotropin receptor, UniProt P16473), a G protein-coupled receptor. OmniPath lists a directed TSHB (P01222) → TSHR

	(P16473) interaction.
Receiver cell	Thyroid follicular cell (thyroid glandular cell in HPA). HPA: TSHR RNA is cell type enriched in thyroid glandular cells of the thyroid gland.
OmniPath	TSHB (P01222) → TSHR (P16473), directed, interaction type post_translational, supported by ligand-receptor resources (LRdb, Cellinker, HPRD, connectomeDB2020, CellTalkDB, HPMR) plus SIGNOR, NetPath and HINT.
STRING	9 proteins (TSHR, TSHB, CGA, GNAS, PRKACA, CREB1, PAX8, TG, TPO), 18 edges; expected 2, PPI enrichment $p = 9.65e-12$. Enriched: thyroid hormone generation (GO:0006590), KEGG hsa04918 (8 of 73, FDR $1.88e-16$), hsa04024 cAMP signaling, WP2032 TSH signaling. Relevant proteins: GNAS, PRKACA, CREB1, PAX8, TG. (See Part D table.)
IntAct	PRKACA and CREB1, record EBI-46451554: anti-bait coimmunoprecipitation, Homo sapiens (HT-29 cells), PMID 37501933, physical association, MI score 0.4. Supports physical association only, not direct binding or phosphorylation.
Final model	Diagram: Figure 5 (Part F), saved as figures/05_final_model.png in the GitHub repository.

5. Questions to Answer in the README or Laboratory Report

1. What sender cell did you choose, and in what tissue or biological context does it act?

- I chose the pituitary thyrotroph. It is found in the anterior pituitary gland and acts in the endocrine system.
2. What signaling molecule did you identify, and what evidence supports its production or presentation by the sender cell?
- The signaling molecule is TSH (thyroid-stimulating hormone), which is made of the CGA and TSHB subunits. In the Human Protein Atlas, TSHB is enriched in the pituitary gland and in pituitary thyrotropes, and it is annotated as secreted into the blood. HPA also shows TSHB RNA in spermatids, so it is not exclusive to the pituitary.
3. What receptor receives the signal, and which receiver cell did you select?
- The receptor is TSHR (UniProt P16473), a G protein-coupled receptor. The receiver cell is the thyroid follicular cell. HPA shows TSHR in thyroid glandular cells.
4. What type of cell-to-cell signaling is represented: paracrine, endocrine, autocrine, or contactdependent?
- Endocrine signaling. TSH is released from the pituitary into the bloodstream and travels to the thyroid, which is far from the sender cell.
5. Which proteins in your STRING network appear most relevant to the receptor-associated response? Explain briefly.
- In my 9-protein network, the most relevant proteins are TSHR, GNAS, PRKACA, CREB1, PAX8 and TG. TSHR is the receptor. GNAS, PRKACA and CREB1 are part of the cAMP/PKA pathway inside the cell. PAX8 and TG are linked to thyroid-specific gene activity and thyroid hormone production. The network had 18 edges while only 2 were expected ($p = 9.65e-12$). STRING edges only show functional association, not direct binding. The enrichment is also partly expected because I chose thyroid-related proteins.
6. What enriched pathway or biological process is consistent with your proposed mechanism?
- The enriched terms were thyroid hormone generation (GO:0006590), thyroid hormone synthesis (hsa04918), cAMP signaling pathway (hsa04024) and WP2032 Thyroid stimulating hormone TSH signaling. These fit my proposed mechanism of TSH binding TSHR, activating cAMP signaling, and leading to thyroid hormone production.
7. What did IntAct show for the molecular interaction you examined? What type of evidence was reported?

- IntAct showed a physical association between PRKACA and CREB1 (EBI-46451554, PMID 37501933). I also found PRKACA and PRKAR1A (EBI-16107177, PMID 24855271) as supporting evidence. I did not find an IntAct record for TSHR with TSHB, CGA or GNAS. This only means no record was found, not that there is no interaction.

8. Which parts of your final model are strongly supported, and which parts remain an inference?

- The strongly supported parts are TSHB expression in the thyrotroph, TSHR expression in the thyroid, the OmniPath TSHB → TSHR ligand-receptor record, and the PRKACA–CREB1 physical association in IntAct. The parts that remain inference are TSHR → GNAS → cAMP → PRKACA and CREB1 → PAX8 → TG. These come from known cAMP/PKA biology and STRING associations, not direct evidence from these databases. They are the dashed arrows in my Figure 5.

9. What cellular response is expected in the receiver cell, and why?

- I expect thyroid hormone synthesis and growth of the thyroid follicular cell. When TSH binds TSHR, the Gs/cAMP/PKA pathway activates CREB1 and thyroid genes such as PAX8 and TG. This agrees with the enriched thyroid hormone and cAMP terms in STRING.

7. Reference Resources

Unique Sender Link: <https://www.proteinatlas.org/ENSG00000134200-TSHB>

Signal/ Ligand: <https://www.proteinatlas.org/ENSG00000134200-TSHB>

Receptor: <https://www.proteinatlas.org/ENSG00000165409-TSHR>

OmniPath: <https://explore.omnipathdb.org/search?q=TSHB,+&tab=interactions&species=9606>

Receiver Cell: <https://www.proteinatlas.org/ENSG00000165409-TSHR>

Protein Network: <https://string->

[db.org/cgi/network?taskId=bH1QFlg0Th6v&sessionId=bIBeScWpzN5T](https://string-db.org/cgi/network?taskId=bH1QFlg0Th6v&sessionId=bIBeScWpzN5T)

Cellular Response: <https://www.proteinatlas.org/ENSG00000042832-TG>

IntAct Validation: <https://www.ebi.ac.uk/intact/search?query=EBI-46451554>

GitHub Repository: <https://github.com/Jayeydd/cell-cell-communication>